

Rajapandiyan Arivazhagan

Senior Generative AI / Applied AI Engineer | LLM Applications | RAG & Agentic Systems

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PROFESSIONAL SUMMARY

Senior Generative AI / Applied AI Engineer with 9+ years of software engineering experience, including 7 years at Zoho Corporation and 5 years in GenAI and agentic systems, specializing in production LLM applications, Retrieval-Augmented Generation (RAG), agentic AI, and Python backend systems. Architected and deployed an enterprise AI support platform adopted by 400+ technicians, combining multi-agent orchestration, hybrid retrieval with reranking, domain fine-tuned LLMs, and custom Model Context Protocol (MCP) servers for governed agent access to SQL databases and internal APIs. Experienced across self-hosted and cloud LLM platforms (Amazon Bedrock, Azure OpenAI, Vertex AI), AI security guardrails, and enterprise-scale automation, with an unusually deep systems background in Windows reverse engineering and patch intelligence.

CORE TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, AI Agents, Model Context Protocol (MCP), MCP Server Development, LLM Fine-Tuning, Prompt Engineering, Tool/Function Calling, Structured Outputs (JSON Schema), Text-to-SQL, Agent Skills / Skills Specification, Context & Memory Management, Multimodal Models (Text, Image), Vision-Language Models (VLMs), Screen Understanding, Computer-Use Agents

Retrieval & Search: Document Ingestion, Intelligent Chunking, Embedding Models, Embedding Generation, Semantic Search, Vector Search, Reranking, Hybrid Retrieval, Knowledge Base Indexing

Frameworks & Models: LangChain, LangGraph, Google ADK, Hugging Face Transformers, PyTorch, OpenAI API, Claude / Anthropic API, vLLM, Ollama, Open WebUI, Langfuse, Self-Hosted / On-Premises LLM Deployment

Cloud AI & MLOps: Amazon Bedrock, Amazon SageMaker AI, Azure OpenAI, Microsoft Foundry, Google Vertex AI, Microsoft 365 Copilot, AWS & Azure AI Deployments, Kubernetes, containerd, CI/CD Pipelines, Databricks, n8n

Databases & Vector Stores: Microsoft SQL Server, SQLite, SQL, ChromaDB, FAISS, Redis

Backend & Infrastructure: Python, Java, JavaScript, FastAPI, SQLAlchemy, REST API Design & Integration, JSON/XML Parsing, Docker, Git, Linux, Windows Server, Streamlit, Plotly, pandas, NumPy

AI Safety & Security: Prompt Injection Mitigation, Sensitive Data Masking (PII, internal identifiers), Input/Output Sanitization, Static Application Security Testing (Semgrep, Bandit), OWASP Top 10, CWE

Enterprise & Endpoint Engineering: Windows Patch Management, Third-Party Application Patching, Software Reverse Engineering, Installer Analysis (EXE, MSI, MSIX), Silent Installation, Software Packaging, Deployment Automation, Detection Logic, CVE Analysis, Endpoint Management, Web Crawling (Selenium, BeautifulSoup, Jsoup)

PROFESSIONAL EXPERIENCE

Member Technical Staff — ManageEngine Endpoint Central, Zoho Corporation

2019 – Present

Technologies: Python, LangChain, LangGraph, Google ADK, MCP, Hugging Face Transformers, PyTorch, OpenAI API, Claude API, Amazon Bedrock, Azure OpenAI, Vertex AI, vLLM, Ollama, Langfuse, Redis, ChromaDB, FAISS, FastAPI, Docker, Kubernetes, CI/CD, Microsoft SQL Server, Selenium, BeautifulSoup, Jsoup, Open WebUI

Generative AI & LLM Engineering

- Enabled 400+ support technicians to access grounded, context-aware troubleshooting guidance by architecting and deploying a production multi-agent AI support platform integrating Zoho Desk APIs, enterprise RAG, MCP-governed SQL and API access, and domain fine-tuned LLMs.
- Engineered an end-to-end enterprise RAG pipeline spanning support tickets, engineering notes, knowledge base articles, and solution repositories, combining intelligent chunking, embedding generation, hybrid retrieval, and reranking to improve grounding quality of technical responses.
- Reduced irrelevant context retrieval by implementing hybrid search and reranking over vector search, improving the relevance of enterprise context supplied to language models.
- Built custom Model Context Protocol (MCP) servers that securely expose Microsoft SQL Server, internal REST APIs, file systems, and enterprise applications to AI agents through governed tool interfaces, enabling agents to retrieve live operational context and execute controlled actions.
- Designed multi-agent architectures with LangChain and LangGraph in which specialized agents interpret ticket context, query internal systems, reason over historical cases, and recommend validated solutions.

- Fine-tuned open-source LLMs on curated internal support data — customer queries, resolved answers, and product-specific troubleshooting logic — improving domain accuracy and reducing generic model responses.
- Evaluated and prototyped hosted LLM platforms — Amazon Bedrock, Azure OpenAI, Google Vertex AI, and SageMaker AI — benchmarking them against self-hosted open models on answer quality, latency, cost, and data residency to select the right deployment model per workload.
- Hardened the platform for enterprise use with prompt injection mitigation, automated masking of PII and internal identifiers before inference, and JSON schema-validated structured outputs for reliable downstream automation.
- Operated self-hosted LLM infrastructure on vLLM, Ollama, Open WebUI, and Docker to keep sensitive enterprise data fully on-premises, exposing AI capabilities as FastAPI REST services consumed by internal tooling, with Langfuse tracing, Redis caching, and CI/CD pipelines for build and release.

Enterprise Automation & Software Intelligence

- Built and maintained enterprise-scale web crawlers in Python (Selenium, BeautifulSoup, Jsoup) that continuously harvest software intelligence — versions, release notes, download URLs, patch metadata, security advisories, and CVE data — across a large portfolio of vendors, handling JavaScript-rendered pages, authenticated sessions, and anti-scraping protections.
- Reverse engineered Windows installers (EXE, MSI, MSIX) through process, registry, file system, and network behavior analysis to determine silent installation parameters, dependencies, and undocumented switches, extending automated patch coverage to applications lacking official silent install support.
- Diagnosed and resolved patch deployment failures through root cause analysis of installer behavior, environment conflicts, and update supersedence, applying deep knowledge of Windows Update architecture — cumulative updates, servicing stack updates (SSU), and the Microsoft Update Catalog — to maintain accurate patch metadata and compliance reporting.

Data Analyst — MicroPrecision

2017 – 2019

- Consolidated production, quality, and process data from multiple operational systems into unified SQL and Python (pandas, NumPy) datasets that became the standard source for recurring engineering and management reporting.
- Automated recurring collection, cleaning, validation, and reporting workflows, replacing manual Excel processes and improving the consistency of operational reporting.
- Investigated process anomalies and quality issues through exploratory data analysis and trend analysis, surfacing root causes that standard reports did not reveal.
- Partnered with engineering, operations, and quality stakeholders to define reporting requirements, validate business logic, and resolve data discrepancies, delivering KPI dashboards used in management reviews.

SELECTED AI PROJECTS

Designed and built internally at Zoho Corporation (ManageEngine).

AI-Powered Support Automation Platform — *Python, LangChain, LangGraph, MCP, RAG, ChromaDB, Ollama, OpenAI API, FastAPI, Docker*
Production multi-agent AI assistant adopted by 400+ support technicians, combining Zoho Desk ticket ingestion, enterprise RAG with reranking, MCP-governed access to SQL databases and internal APIs, fine-tuned domain models, and prompt injection and data-masking guardrails on self-hosted infrastructure.

QueryMind — GenAI Text-to-SQL Analytics Assistant — *Python, LangChain, SQLAlchemy, RAG, LLMs, Streamlit, Plotly*
Converts natural-language questions into SQL and executes them against relational databases using schema-aware, RAG-grounded query generation. Interactive Streamlit and Plotly visualizations let non-technical users explore business data without writing SQL.

VulnGuard AI — LLM-Based Source Code Vulnerability Analyzer — *Python, FastAPI, LangChain, FAISS, Streamlit, Semgrep, Bandit, RAG*
Detects SQL injection, XSS, command injection, insecure authentication, hardcoded credentials, and unsafe dependencies by combining static analysis with LLM reasoning. Integrates Semgrep and Bandit findings with LLM-generated explanations, severity assessment, and remediation guidance, grounded in OWASP and CWE knowledge indexed in a FAISS vector store.

Vision-Based Windows Troubleshooting Agent (early-stage prototype) — *Python, GPT-4o Vision, Qwen-VL, Vision-Language Models, Agent Loop, Windows Automation*

Exploratory autonomous endpoint troubleshooting agent that interprets Windows screen state visually, navigates the GUI, and performs diagnostic and remediation steps without human interaction. Uses GPT-4o vision and Qwen-VL for screen understanding and UI element grounding, with a plan-act-verify agent loop that executes actions and confirms the resulting screen state before proceeding.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering — Kamaraj College of Engineering and Technology, 2017 | **Languages:** English, Tamil